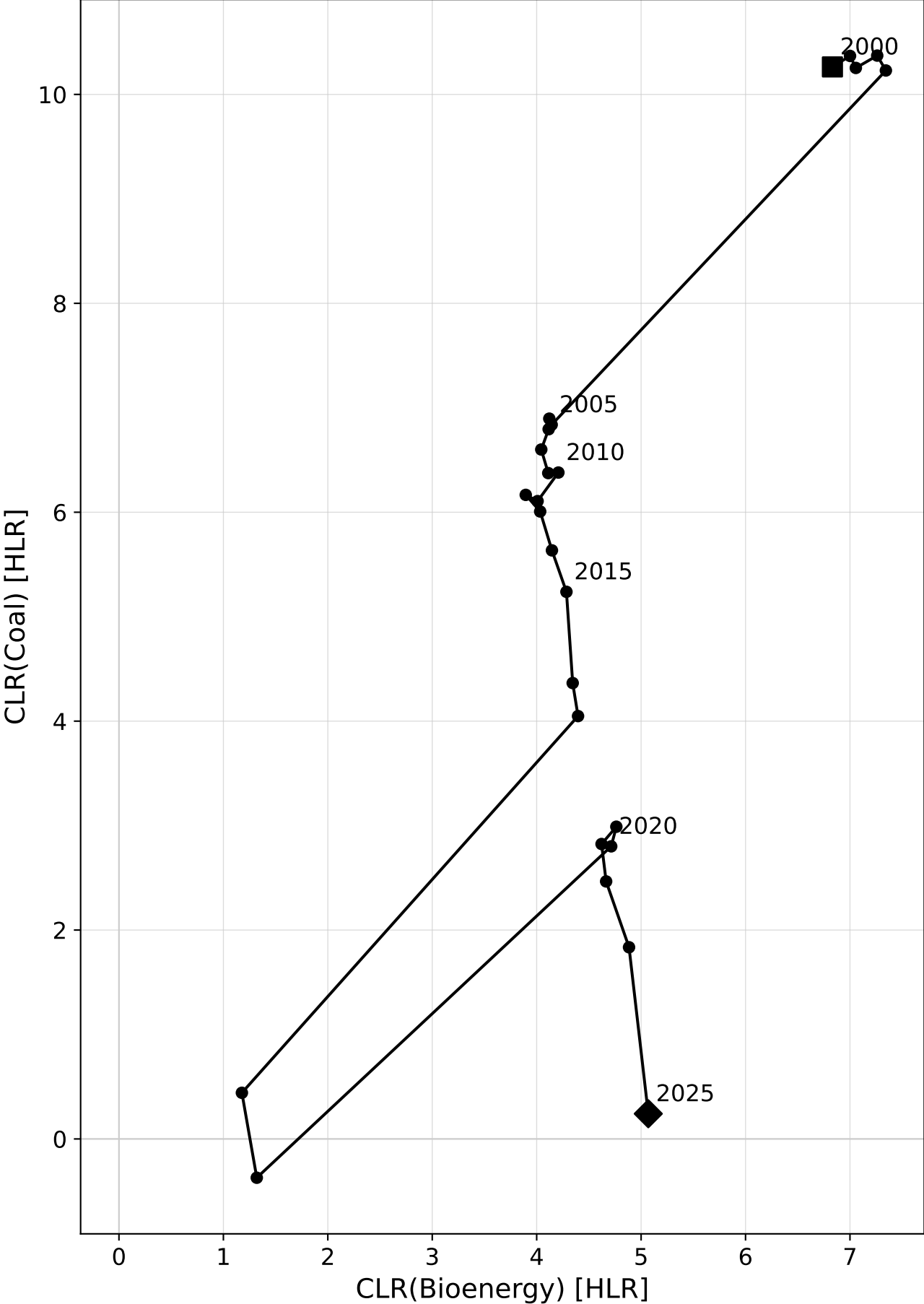
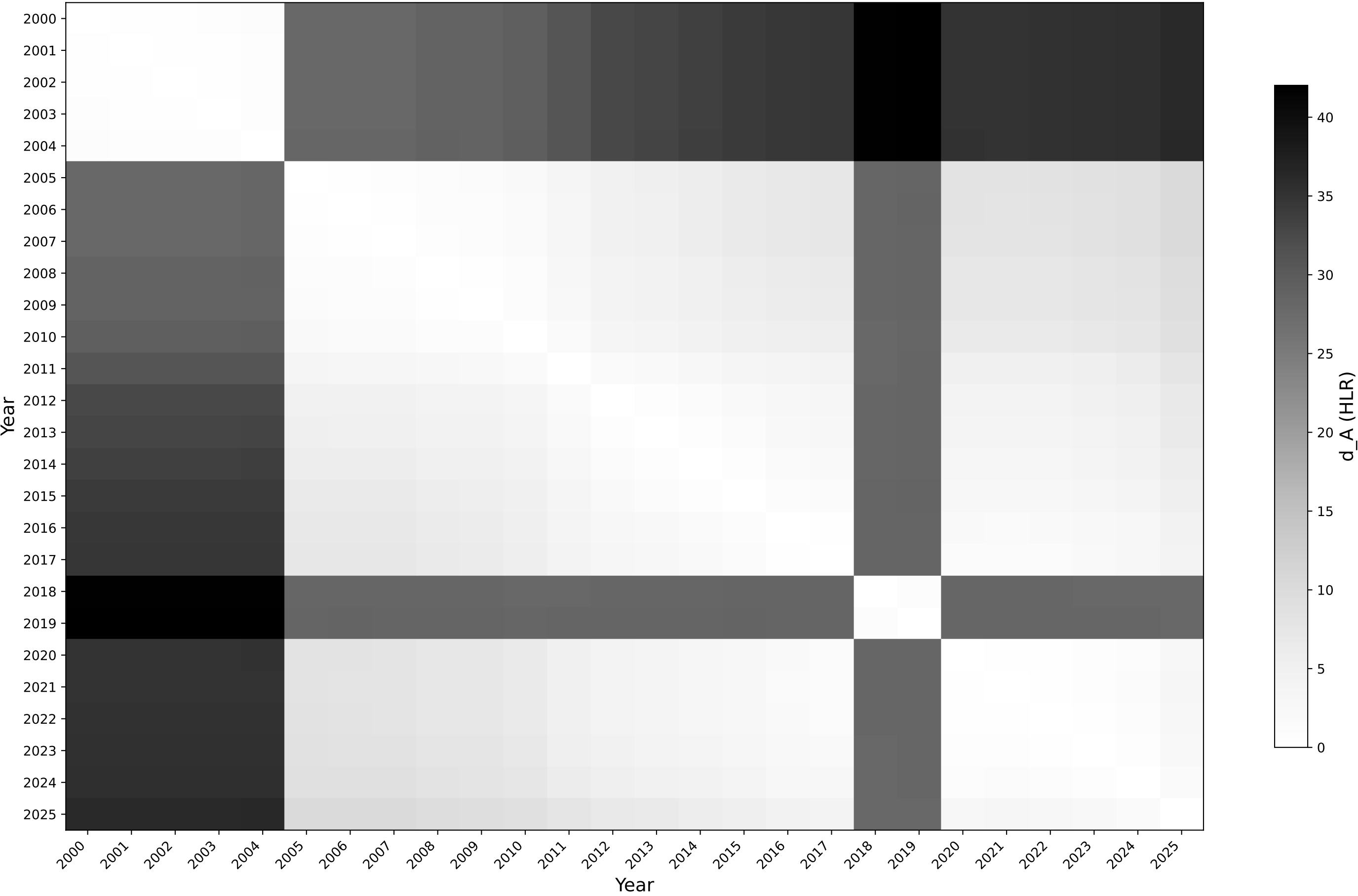


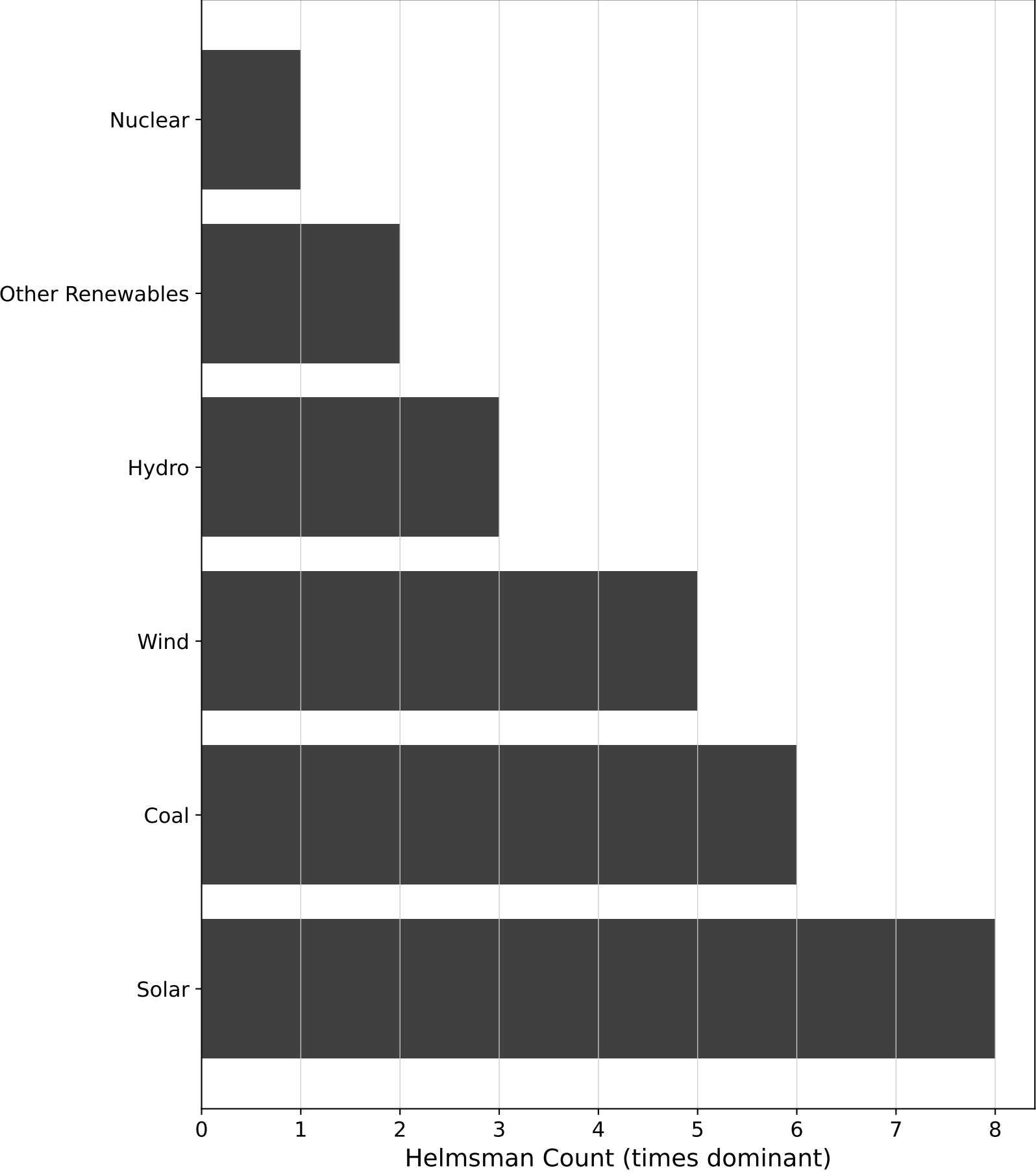
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



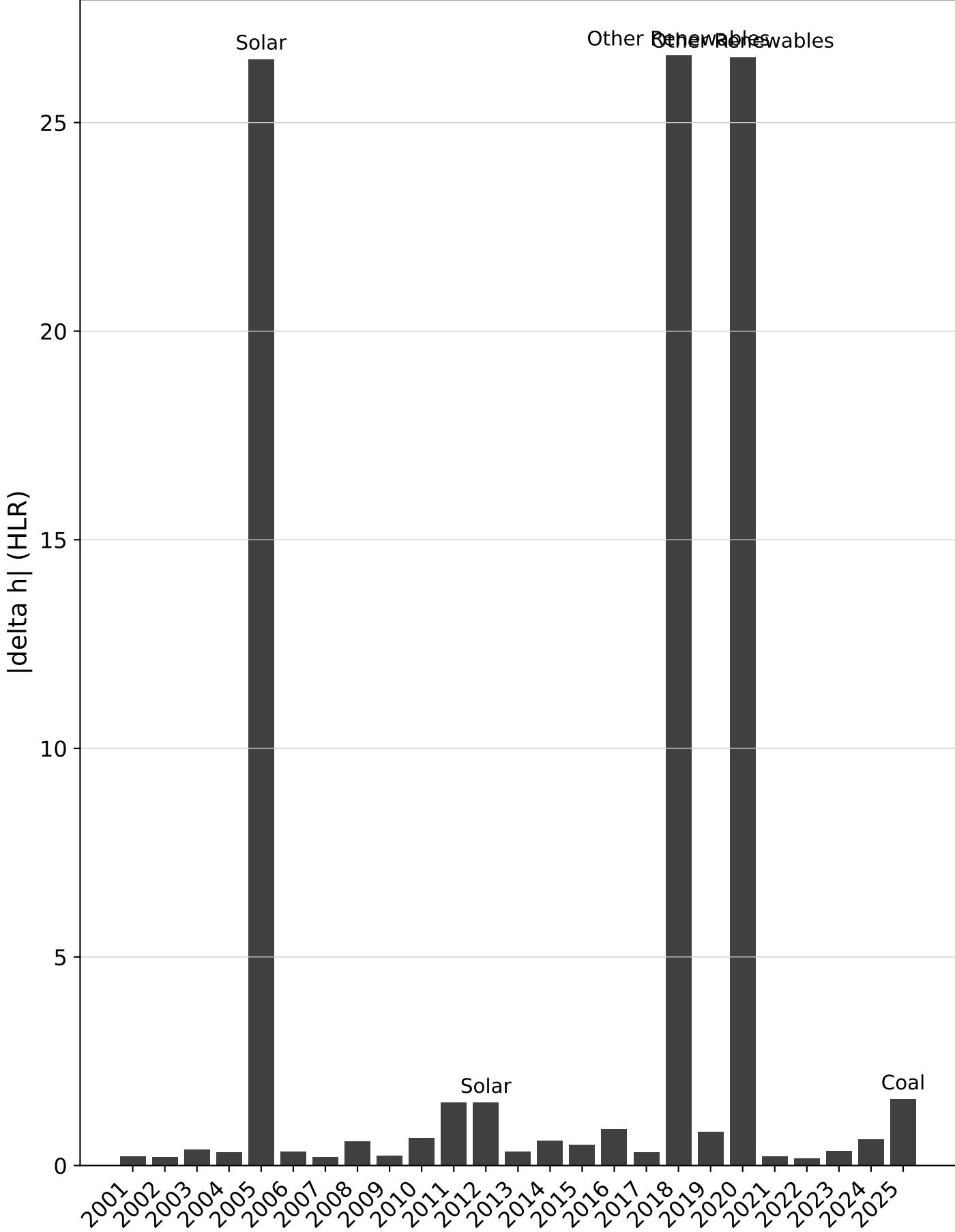
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



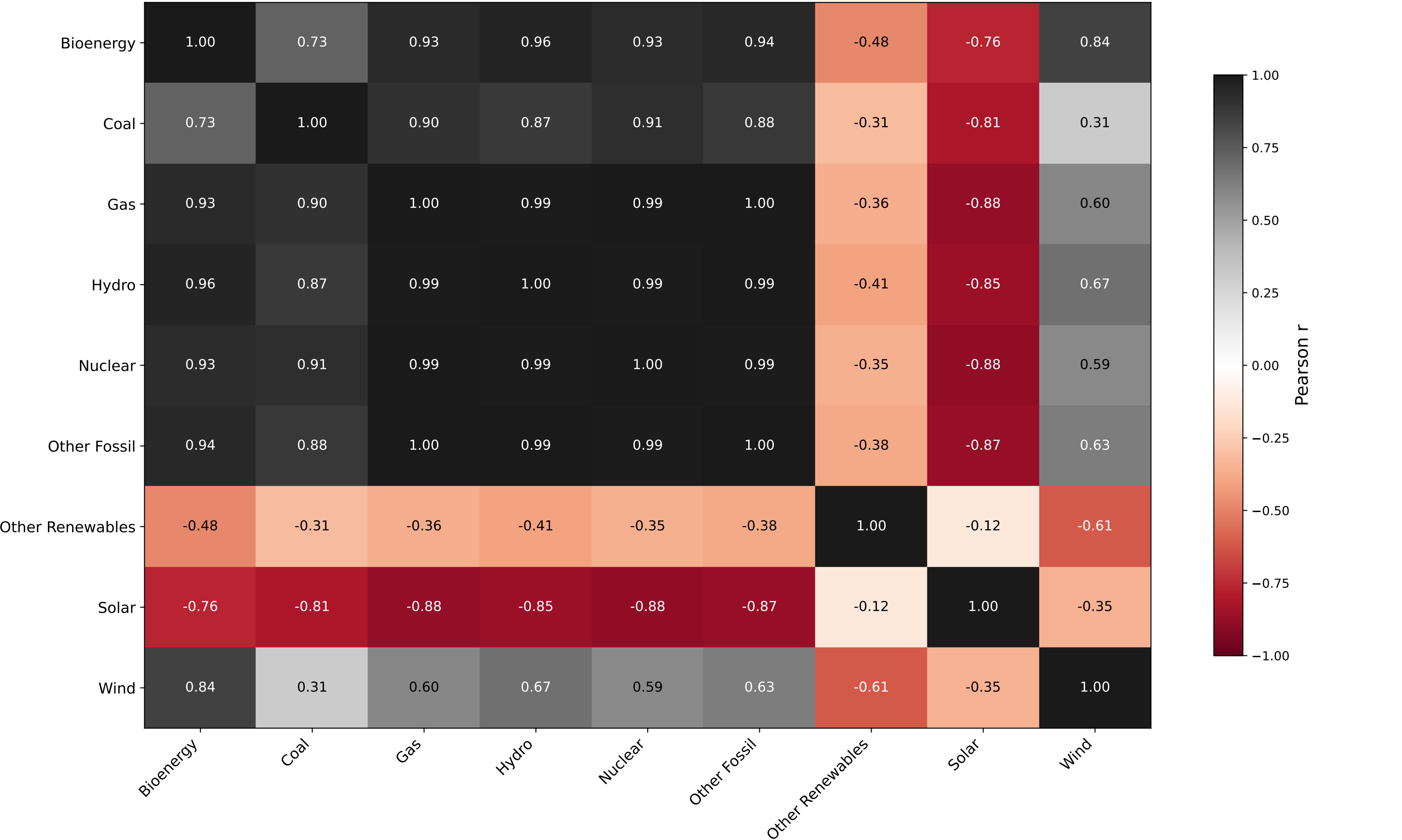
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2004	2019	42.0223	47.2	Solar
2	2002	2019	41.9369	47.4	Solar
3	2003	2019	41.9196	47.3	Solar
4	2000	2019	41.9006	47.5	Solar
5	2001	2019	41.8750	47.4	Solar
6	2004	2018	41.8489	45.7	Solar
7	2002	2018	41.7611	45.8	Solar
8	2003	2018	41.7407	45.7	Solar
9	2000	2018	41.7240	46.0	Solar
10	2001	2018	41.6954	45.9	Solar
11	2004	2025	36.2846	49.8	Solar
12	2002	2025	36.2613	49.9	Solar
13	2000	2025	36.2591	49.9	Solar
14	2003	2025	36.2581	49.9	Solar
15	2001	2025	36.2497	49.9	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2004	2019	2025	501.5179	1.50
2	2004	2018	2025	501.2999	1.49
3	2002	2019	2025	500.9550	1.50
4	2003	2019	2025	500.8574	1.50
5	2000	2019	2025	500.8000	1.49
6	2002	2018	2025	500.7174	1.49
7	2003	2018	2025	500.6060	1.49
8	2001	2019	2025	500.6035	1.49
9	2000	2018	2025	500.5544	1.49
10	2001	2018	2025	500.3433	1.49

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4037	Hs-3	1560.2	0.00	0.0000	---	1000000000000000
2001	0.4099	Hs-3	1557.0	0.42	0.3465	Hydro	1000000000000000
2002	0.4066	Hs-3	1561.6	0.35	0.2975	Wind	1000000000000000
2003	0.4110	Hs-3	1558.5	0.55	0.4536	Hydro	1000000000000000
2004	0.4035	Hs-3	1567.4	0.66	0.5488	Wind	1000000000000000
2005	0.3881	Hs-3	905.5	36.63	28.1325	Solar	1000000000000000
2006	0.3824	Hs-3	907.5	0.65	0.4070	Wind	1000000000000000
2007	0.3931	Hs-3	907.4	0.61	0.3767	Wind	1000000000000000
2008	0.3922	Hs-3	905.0	1.17	0.7295	Solar	1000000000000000
2009	0.3654	Hs-3	908.6	0.70	0.4341	Nuclear	1000000000000000
2010	0.3807	Hs-3	902.7	1.34	0.8341	Solar	1000000000000000
2011	0.3375	Hs-3	903.4	2.78	1.7268	Solar	1000000000000000
2012	0.3153	Hs-3	905.6	2.72	1.6885	Solar	1000000000000000
2013	0.2873	Hs-2	908.8	0.91	0.5695	Solar	1000000000000000
2014	0.2457	Hs-2	914.6	1.22	0.7609	Solar	1000000000000000
2015	0.2061	Hs-2	919.4	1.10	0.6902	Solar	1000000000000000
2016	0.2499	Hs-2	916.8	1.67	1.0458	Coal	1000000000000000
2017	0.2424	Hs-2	917.3	0.73	0.4586	Coal	1000000000000000
2018	0.2384	Hs-2	63.1	16.61	28.2360	Other Renewables	33376
2019	0.2580	Hs-2	63.1	6.43	0.8914	Coal	32719
2020	0.2468	Hs-2	915.4	17.72	28.1817	Other Renewables	1000000000000000
2021	0.2555	Hs-2	915.0	0.59	0.3689	Hydro	1000000000000000
2022	0.2593	Hs-2	914.4	0.49	0.3091	Wind	1000000000000000
2023	0.2507	Hs-2	914.9	0.70	0.4445	Coal	1000000000000000
2024	0.2392	Hs-2	915.2	1.14	0.7109	Coal	1000000000000000
2025	0.2455	Hs-2	914.6	2.77	1.7159	Coal	1000000000000000